

Sam Goodwin

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Profile

Computer Science graduate (Bridgewater State University, 2025, Magna Cum Laude, GPA 3.8) with hands-on experience building software across the full stack in Java, Python, Go, and C++. Developed REST APIs, relational/embedded data layers, interactive GUI/CLI interfaces, concurrent systems, and clean modular architecture with an emphasis on unit/integration testing. Integrates AI coding assistants (Claude, Grok, GitHub Copilot) directly into daily development, using chat within IDE and autocomplete alongside browser-based tools to accelerate implementation while critically reviewing all generated code for correctness, security, performance, and maintainability. Certified by HackerRank in SQL, Go, Java, and Python.

Education

Bachelor of Science in Computer Science

Bridgewater State University, 2025

- Magna Cum Laude (GPA: 3.8)
- Dean's List (All Semesters)

Skills

Languages: Java, Python, Go, C++, JavaScript, SQL, Bash/Shell

Databases: SQL, SQLite

AI Tools: Claude, Grok, GitHub Copilot, Gemini (IDE chat, autocomplete, browser-based generation and review)

Libraries/Frameworks: OpenCV, face_recognition, MediaPipe, DJI Tello, Excelize, SQLite3, Pyne, JavaFX, Maven, React

Core Fundamentals: Data Structures, Algorithms, Object-Oriented Design, Concurrency and Multithreading, Unit/Integration Testing, GUI/CLI Development, Computer Vision, IoT Systems, REST API Integration, Data Visualization, Mapping and Geocoding

Tools: Git, GitHub, Docker, VS Code, Cursor, Arduino, Raspberry Pi, PlatformIO, JetBrains

Operating Systems: Linux, Windows

Certifications

- **HackerRank:** SQL (Advanced, Intermediate, Basic), Go (Intermediate and Basic), Java (Basic), Python (Basic) Skills Certifications

Projects

Job Visualizer

github.com/goodwin-sam/job-visualizer

- Built a cross-platform Go desktop app (GUI + CLI) for importing, processing, and visualizing job data from Excel files, backed by a SQLite database.
- Designed an interactive Fyne GUI with advanced filtering (keyword, location, salary, and remote status), OpenStreetMap Nominatim API mapping with local geocoding cache, and interactive data visualization charts.
- Containerized the application with Docker and configured a GitHub Actions CI pipeline for automated builds, with portable release binaries (Applmage for Linux, .exe for Windows).
- Maintained clean architecture through cmd/pkg separation, conventional commits, comprehensive testing, and detailed documentation.

Concurrent File Processor

github.com/goodwin-sam/concurrent-file-processor

- Developed a cross-platform Java desktop application for concurrent multi-file text analysis, computing word frequencies, character/line counts, and aggregate file statistics via GUI or headless CLI.
- Implemented custom work-stealing thread pools for load-balanced parallel processing.
- Designed a thread-safe, modular architecture with clear package separation, a JavaFX multi-window GUI, and portable executables for Windows and Linux.

Smart Tello Follower

github.com/goodwin-sam/smart-tello-follower

- Built a real-time computer vision system in Python that streams live video from a DJI Tello drone, performs real-time face recognition (OpenCV and face_recognition) to autonomously track a target.
- Implemented hand gesture flight control via MediaPipe, with clean modular separation between drone control, face recognition, and gesture recognition subsystems.
- Implemented a centralized configuration module for tunable detection thresholds and flight parameters, with clean dependency management.

Environment Sensor Hub

github.com/goodwin-sam/enviro-sensor-hub

- Built an Arduino-based C++ environmental monitoring system that reads multiple sensors (temperature, humidity, water level, ambient light) to control a fan, buzzer, and a 16x2 LCD.
- Implemented a PWM fan speed control across temperature tiers and automatic LCD backlight adjustment based on ambient light.
- Organized the codebase into a modular structure (sensors, display, serial utilities) with configurable buzzer alarms and all thresholds/pin assignments centralized in a config header for easy tuning.

Experience

Property Manager

Private Landlord, Massachusetts

June 2020 – Present

- Managed day to day operations for multiple residential rental properties, consistently responding to tenant requests same-day or next-day. Including on-call night/weekend work for critical issues (e.g., no heat or hot water).
- Performed hands-on troubleshooting and repairs for plumbing, electrical, appliance, and general maintenance issues while managing lawn care and snow removal.
- Handled rent collection, accurate financial record-keeping, and administrative tasks with strong attention to detail.
- Coordinated and oversaw external contractors, ensuring timely completion and high-quality work.
- Demonstrated strong organization, reliability, and problem-solving in a fast-paced high-demand environment through consistent documentation, scheduling and following up.

Teaching Assistant/Tutor

Curry College, Milton Massachusetts

January 2019 – May 2020

- Delivered one-on-one and small-group tutoring sessions in Chemistry and Biology for undergraduate students.
- Broke down complex scientific concepts, experimental processes, and problem-solving techniques, helping students improve critical thinking and academic performance.
- Held regular office hours and review sessions to address conceptual gaps, review homework/lab reports, and provide constructive feedback.
- Supported faculty by preparing study materials, facilitating group discussions, and assisting students with rigorous science coursework.